Partial cross-location generalization of a CA1 wall-related population contrast

An exploratory longitudinal reanalysis

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Abstract

Environmental boundaries strongly influence hippocampal place fields, but it is unresolved whether wall-related remapping generalizes across physical locations. We reanalysed longitudinal calcium-imaging data from 5,413 dorsal CA1 neurons in seven mice across repeated exposure cycles containing a familiar square and nine internal-wall geometries. A cell-resolved wall-minus-open contrast estimated at one location predicted later held-out responses at the same location (mean correlation advantage 0.184; seven of seven animals). A single-tile counterfactual gave the same result (0.225; seven of seven), including after adjustment for speed, heading, time, occupancy, and spatial sampling. A contrast estimated at a non-overlapping source seam predicted a later wall response at a target seam 25 cm away ($\overline{\rho} = 0.148$; seven of seven animals) and remained positive after behavioral adjustment ($\overline{\rho} = 0.119$). Cross-location prediction was weaker than a query-matched exact-location benchmark ($\overline{\rho} = 0.230$; shifted-minus-exact $= -0.082$), demonstrating partial rather than complete generalization. The correctly oriented source also exceeded an empirical equal-midpoint wrong-orientation control by 0.101 in all seven animals. A strict same-session reflected-open control favored the wall target by 0.114 in all five eligible animals, although raw-rate and fully distance-stratified comparisons were less consistent. This pattern shows that a CA1 wall-related population contrast generalizes beyond its original coordinate while retaining strong spatial dependence. The exploratory, fixed-order design does not identify whether this phenomenon reflects a wall-specific mechanism or more general spatial deformation.

Keywords: hippocampus; CA1; place cells; boundaries; population code; transfer; remapping; cognitive map

Significance statement

A cell-by-cell CA1 response contrast associated with a wall at one location predicted a later wall response somewhere else in all seven animals. The prediction remained after behavioral adjustment but was weaker than at the original location. Spatial controls support relative wall specificity while leaving some contribution from smooth place-field structure unresolved. CA1 remapping therefore shows graded generalization: related across locations, but still anchored to place.

1 Introduction

Place cells relate an animal's position to a distributed hippocampal activity pattern, yet the map is not a passive image of Euclidean space. Place fields are anchored by environmental geometry, shift when boundaries move, and can appear at characteristic distances and directions from walls (O'Keefe and Burgess, 1996; Hartley et al., 2000; Barry et al., 2006). Boundary-responsive inputs provide one mechanistic route by which local geometry could organize hippocampal representations (Lever et al., 2009). More generally, current accounts treat hippocampal-entorhinal codes as structured representations whose parts may be recombined or generalized across experiences (Stachenfeld et al., 2017; Wernle et al., 2018; Bakermans et al., 2025).

A wall-related response can recur at the same coordinate despite changes elsewhere in the environment. We call this *exact-location reuse*: a local feature of an otherwise changing map remains stable. *Cross-location generalization* means something different: the cell-resolved contrast evoked by a wall at one seam predicts the response to a wall at another seam. Ordinary place-field stability and boundary-vector-like firing do not guarantee this cell-by-cell prediction across coordinates.

Comparing cross-location prediction with exact-location reuse separates three possibilities. Similar values would imply little dependence on coordinate; a smaller cross-location value would imply partial generalization; and a value near zero would imply that the response is rebuilt at each location. Mean rate maps can look similar in all three cases, so the distinction requires tracking the same cells across sessions.

The dataset of Lee et al. (2025) contains the needed geometry and longitudinal cell registration. Seven mice were imaged in dorsal CA1 while foraging in a familiar square and in nine environments made by placing internal walls on a common spatial lattice. Repeated exposure cycles provided registered cells, recurring local seams, and later held-out sessions. The design was not created for this analysis, so environment order is fixed rather than randomized. We measured recurrence at the same seam, isolated a clean single-tile contrast, and tested whether a contrast estimated at one seam predicted the response at a neighboring, non-overlapping seam.

All three predictions were positive, with a clear gradient. Exact-location prediction was strongest; cross-location prediction was smaller but positive in every animal. A wall-related CA1 population pattern can therefore recur at a different coordinate without becoming independent of its original location.

2 Results

2.1 A held-out test of cell-resolved wall contrasts

Each candidate wall location was represented as an oriented seam between two adjacent lattice tiles. For each registered cell, environment, and seam, we estimated activity in a narrow local strip and subtracted the corresponding estimate from a bracketing familiar-square session. This square residual reduced stable place preference while retaining the environment-specific local change.

Training geometries supplied two cell-wise profiles at each seam: a mean residual when the seam was blocked by a wall and a mean residual when it was open. Their difference was the estimated wall-related contrast. The neural outcome from the later target geometry was never used to form that source contrast. Prediction was scored by a rank correlation across held-out registered cells. We report animal-level means and sign counts rather than treating cells or queries as independent biological replicates.

2.2 A local wall-open contrast recurs across global geometries

For exact-location reuse, the broader wall context was matched between training and held-out comparisons. Across 252 held-out predictions, the wall-derived profile correlated more strongly with later wall responses than did the open-derived profile. The mean wall-minus-open correlation advantage was 0.184 and was positive in all seven animals (Fig. 2A). The sign was also positive in all animals when horizontal and vertical seams were evaluated separately.

This is more specific than a population-average increase near walls. The predictor is indexed by registered cell identity: cells with positive or negative values in the estimated local contrast tended to retain those signs when the same seam appeared in another global geometry. Because the seam was unchanged, this result addresses recurrence at one location, not generalization across locations.

2.3 A one-tile counterfactual sharpens exact-location reuse

Two environments in the sequence, denoted u and o, differed only in whether one lattice tile was blocked. Their contrast therefore isolates one local wall state more cleanly than comparisons among complex layouts. We estimated the $u-o$ cell-wise contrast at that focal seam and tested it in held-out i, l, and t geometries that independently instantiated wall or open states there.

Across 46 held-out predictions, the estimated vector correlated with wall outcomes ($\overline{\rho} = 0.197$) but not open outcomes ($\overline{\rho} = -0.028$), giving an advantage of 0.225, positive in all seven animals (Fig. 2A). The raw-rate version gave a similar advantage (0.243). A model that adjusted local bin estimates for speed, heading, elapsed time, and occupancy retained the result, both with time included (0.225) and omitted (0.229).

The effect was more evident in the nearest independently eligible distance band, 2.5–7.5 cm from the seam (mean advantage 0.233, seven of seven animals), than in a 17.5–22.5 cm band (0.069, six of seven; Fig. 2D). These bands had separately defined supports, making the near–far comparison descriptive rather than paired.

The sequence was fixed: u always followed o. A generic same-seam, unchanged-state later-minus-earlier placebo was opposite in sign (mean -0.042, zero of seven positive), whereas the actual $u - o$ contrast was positive (mean 0.195, seven of seven) and remained so after the nuisance adjustment (mean 0.212). This argues against a generic drift explanation, but it cannot remove an environment-specific order effect because wall state and order were never randomized.

2.4 A wall-related population contrast generalizes across locations

For each eligible target, a wall-minus-open vector was estimated at a distinct source seam with the same signed wall normal. The source and target seams were 25 cm apart, their local analysis bins did not overlap, and source training sessions were disjoint from the target session. The nearest source-target bins were 5 cm apart for tangential translations and 15 cm apart for normal-axis translations. The test used a separate multi-geometry estimator rather than the $u - o$ contrast.

The translated source vector predicted the later target-wall residual with mean correlation 0.148, positive in all seven animals (Fig. 2B). It also outperformed a source-open predictor by 0.152, and was more aligned with the target-wall response than with a training-derived target-open profile by 0.216, again positive in every animal. This cell-by-cell result is more specific than elevated activity at two wall locations: the signed contrast estimated

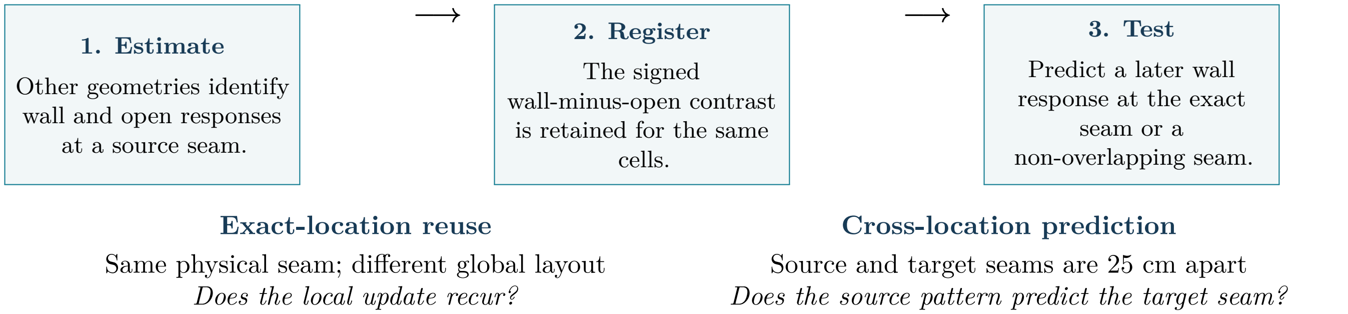

Figure 1: Analysis logic. All tests use registered-cell response vectors estimated from training geometries and evaluated on later held-out neural outcomes. Exact reuse and cross-location prediction are separate estimators: the cross-location result is not a transported version of the single-tile counterfactual.

at one coordinate predicted the response at another.

The exact-location benchmark was higher. On the same queries, cells, and support, it predicted the target response with mean correlation 0.230. Cross-location prediction was lower by 0.082, and only one of seven animals favored the shifted predictor. The response pattern generalizes, but remains strongly tied to its original coordinate.

2.5 Behavioral adjustment and spatial controls

Because animals may run differently near internal barriers, we re-estimated local neural response maps with covariates for speed, first and second circular harmonics of heading, elapsed time, and spatial sampling. The behavior-adjusted cross-location prediction remained positive in all seven animals ($\overline{\rho} = 0.119$). It correlated 0.172 more strongly with the target wall than with a training-derived target-open profile, and it exceeded a source-open predictor by 0.138; both differences were positive in all seven animals. Omitting elapsed time changed these values by less than 0.003.

An empirical spatial control replaced the correctly oriented source with source seams at the same 25 cm target-midpoint distance but with an incorrect signed wall normal. It retains the observed neural maps, registered-cell identities, target response, and source-target lag while breaking the correct wall relation. The correct-minus-incorrect advantage was 0.101, positive in all seven animals. Complete enumeration of the 2^7 animal-level sign assignments placed the observed mean in the upper 1/128 of that descriptive null. The result remained positive in every leave-one-animal-out summary.

This pooled comparison does not eliminate spatial continuity. When restricted to tangential translations with both a 25 cm midpoint distance and a 5 cm nearest-strip lag, the correct-orientation advantage was only 0.040, positive in three of six animals (sign-flip tail fraction 0.25). Both correctly and incorrectly oriented tangential sources predicted the target. The pooled effect is therefore not an arbitrary equal-midpoint match. The most tightly distance-matched stratum, however, does not support a general orientation-specific rule.

Generic nearby place-field structure could still be shared by the source and target strips. A stricter within-session control was specified before its neural endpoint was inspected. For each wall target, a mirror-open seam in the same target session was matched for translation distance, axis, signed normal, relative bins, and registered cells. The geometry permits this match only for tangential translations. Although 104 labeled configurations were feasible, paired-support gates retained 72 triplets in five animals; two animals had no admissible control.

On this restricted subset, the globally demeaned source vector favored the wall target over the mirror-open target by 0.114, positive in all five eligible animals (Fig. 2C). Absolute source-to-wall correlation was positive in four of five animals ($\overline{\rho} = 0.068$), and a direct within-session wall-minus-open difference-vector score was positive in four of five ($\overline{\rho} = 0.077$). The raw-rate primary was positive in only three of five. The exact animal-level sign-flip tail fraction for the demeaned advantage was 1/32, and every leave-one-animal-out mean remained positive. In the supported demeaned tangential cases, the source contrast distinguished a wall target from an equally displaced open location. This control does not cover all seven animals, normal-axis translations, or raw-rate specificity.

A coherent source-cell identity permutation further showed descriptive upper-tail fractions below 0.01 in five animals and values of 0.069 and 0.060 in the other two. Registered-cell correspondence therefore contributes to the prediction, although the effect is not uniformly strong. Additional exploratory analyses of direction, composition, experience, and a former-barrier endpoint are reported in Supplementary Section 5 rather than treated as tests of the central phenomenon.

Across animal-level comparisons, cross-location prediction was consistently positive but lower than exact-location prediction. Spatial specificity was clear in the pooled orientation and mirror-open comparisons, but not in the strict tangential match (Fig. 3).

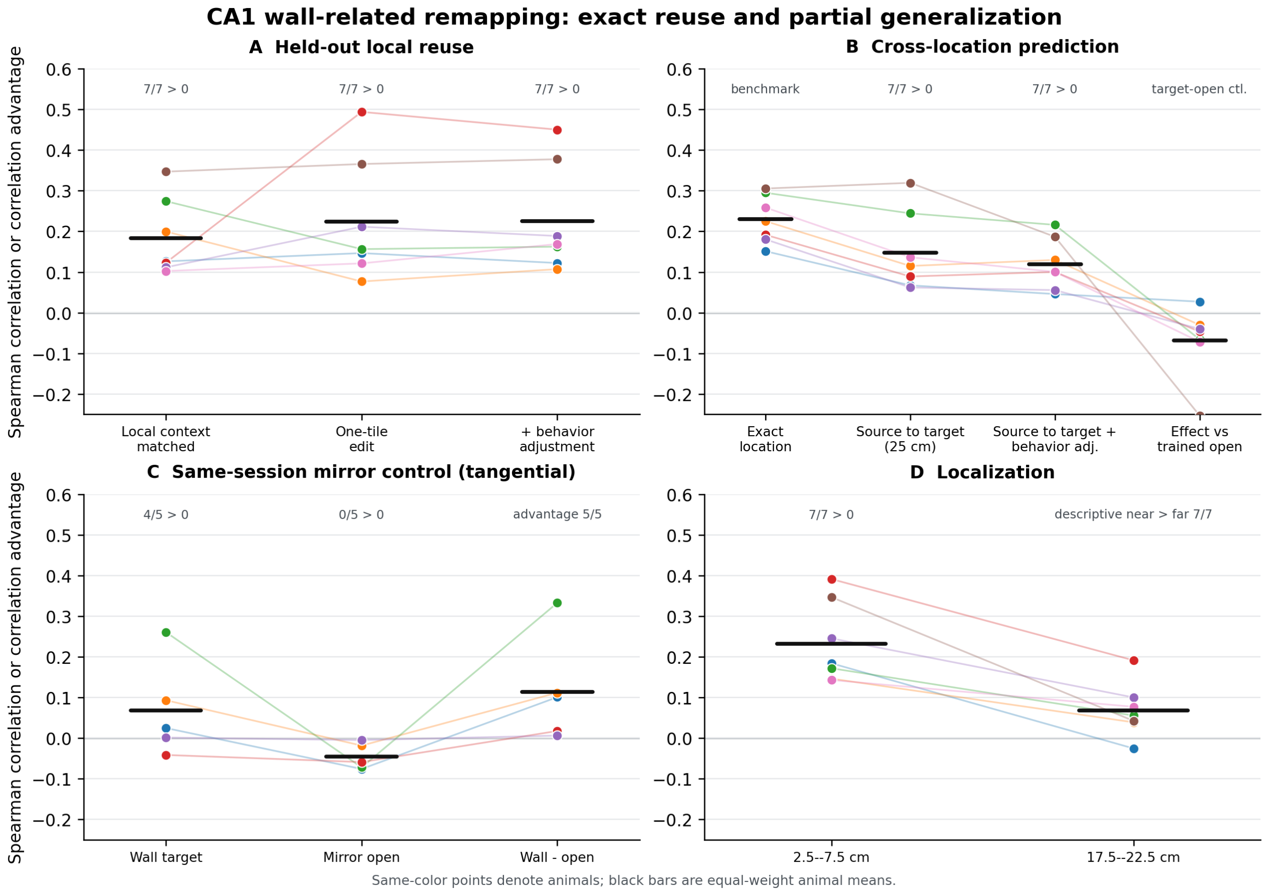

Figure 2: Exact reuse, cross-location prediction, and key controls. A, same-seam wall-minus-open prediction under three estimators. **B**, matched exact-location and source-to-target predictions, including behavioral adjustment and the target-open comparison. **C**, wall and mirror-open targets within the same session; five animals had paired support. **D**, descriptive near and far bands with separate eligibility. Points are animal means; bars are across-animal means.

Table 1: Evidence for partial cross-location generalization. Values are means across animal-level estimates. Sign counts are descriptive consistency summaries.

Test	Mean	Sign	Supported interpretation
Context-matched exact seam	0.184	7/7	Local wall-open population contrast recurs across layouts.
Single-tile exact seam	0.225	7/7	Exact-location reuse remains in a cleaner local counterfactual.
Behavior-adjusted single tile	0.225	7/7	Measured speed, heading, time, and sampling do not explain reuse.
Cross-location source effect	0.148	7/7	A cell-resolved wall contrast predicts a different location.
Query-matched exact effect	0.230	7/7	Same-location prediction is stronger.
Behavior-adjusted cross-location effect	0.119	7/7	The cross-location result remains after behavioral adjustment.
Correct minus wrong orientation at 25 cm	0.101	7/7	The pooled effect exceeds an equal-midpoint empirical spatial null.
Mirror-open advantage	0.114	5/5	Demeaned tangential subset favors a wall over an equally displaced open seam.

3 Discussion

A cell-resolved wall-minus-open response estimated from one group of geometries predicted a later response at the same seam. The same kind of contrast, estimated at a different seam, also predicted the later target-wall response. This cross-location prediction was positive in every animal, remained after adjustment for measured behavior, and was weaker than the matched exact-location prediction. Local boundary-related remapping is related across coordinates but remains anchored to place.

Boundaries are known to influence place fields. Boundary-vector models explain how fields can arise from inputs tuned to distance and direction from boundaries

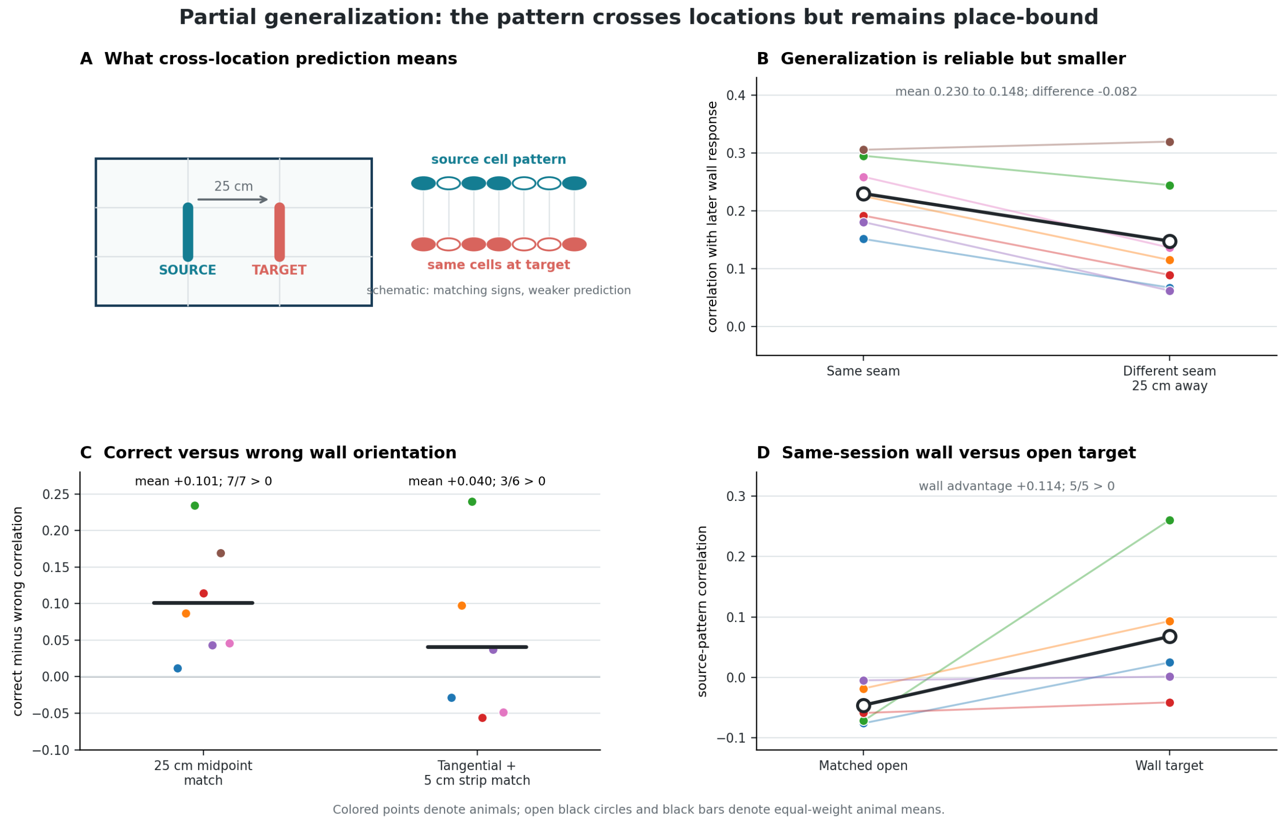

Figure 3: The central result and its spatial limits. **A**, schematic of a source wall-related cell pattern tested at a target seam 25 cm away. Filled and open circles denote schematic signed cell responses, not additional measurements. **B**, paired exact-location and cross-location correlations in seven animals. **C**, correct-minus-wrong orientation effects for the pooled 25 cm control and the strictly matched tangential subset. **D**, source-pattern correlations with same-session mirror-open and wall targets in five eligible animals. Quantitative panels show globally demeaned animal-level estimates.

(Hartley et al., 2000; Barry et al., 2006), and barrier manipulations have shown local field formation, duplication, and geometry-dependent remapping (Rivard et al., 2004; Lee et al., 2025). Those results make recurrence plausible but do not test whether a signed vector of responses, indexed by the identities of longitudinally registered CA1 cells, generalizes from one physical seam to another under held-out evaluation.

The lower cross-location value rules out both extremes. The response is not rebuilt independently at every seam, but most of its predictive structure remains tied to the original location. This is partial generalization and does not require a location-invariant neural operator.

Several alternative explanations remain. First, smooth or repeated place fields can correlate at nearby seams. The non-overlapping strips, square residuals, target-open specificity, equal-midpoint wrong-orientation null, and mirror-open control constrain this account. They do not eliminate it: the fully distance-matched tangential orientation contrast was inconsistent, and the mirror control was limited to five animals and demeaned tangential translations. Second, unmeasured behavior such as attention, wall inspection, acceleration, or route choice could contribute despite adjustment for measured kinematics. Third, the fixed environment sequence leaves environment-specific order confounded with the single-tile contrast. Fourth, registration and activity thresholds select cells stable enough to be observed across sessions. Fifth, the identity permutation is not an ideal exchangeability test for correlated place-cell populations.

Exploratory analyses of composition, experience, and a former-barrier endpoint were underpowered or unstable and appear in the Appendix. They do not support claims about hippocampal composition, learning, or innateness.

A counterbalanced test would use a symmetric lattice arena with four equivalent candidate wall seams. Source and target locations, wall orientation, insertion or removal, and session order would be randomized within animal. Exact reuse, cross-location prediction, and a distance-matched open target could then be measured in the same later session. This design separates wall-specific generalization from smooth spatial deformation and removes the fixed-order confound. The observed values provide design targets: an exact-location correlation near 0.23, a cross-location correlation near 0.12–0.15, and a difference near 0.08.

4 Materials and methods

4.1 Dataset and analysis cohort

We reanalysed the public longitudinal dorsal-CA1 calcium-imaging dataset of Lee et al. (2025, 2024). The tracked analysis cohort contains seven mice, 5,413 neurons, 207 sessions, a familiar outer square, and nine internal-wall geometries arranged on a common lattice. Cell identities were registered across sessions in the source dataset. The original study describes surgery, imaging, behavior, preprocessing, registration, and animal procedures. No new animal experiments were performed.

4.2 Spatial representation and oriented seams

The arena was expressed in a common allocentric coordinate system. A candidate internal-wall location was represented as an oriented seam $s = (p \to q)$ between adjacent lattice tiles, with a signed normal from tile p to tile q. The binary state $z_{e,s}$ indicated whether environment e blocked that adjacency. Local strips were partitioned into 5 cm spatial bins at defined distances from the seam. Analyses used only bins and registered cells satisfying the relevant occupancy, activity, and common-support gates. Source and target strips in the cross-location analysis shared no 5 cm bin.

4.3 Local response vectors

For registered cell i, environment e, and seam s, let $r_{i,e,s}$ be activity averaged over the eligible local bins. Let $b(e)$ denote the paired familiar-square session. We subtract that square baseline and collect the changes across the n shared cells:

$$\begin{aligned} \delta_{i,e,s} &= r_{i,e,s} - r_{i,b(e),s}, \\ \boldsymbol{\delta}_{e,s} &= \left(\delta_{1,e,s}, \ldots, \delta_{n,e,s}\right)^{\mathsf{T}}. \end{aligned} \tag{1}$$

For globally demeaned analyses, the vector mean was removed before correlation. The resulting score depends on relative cell-specific changes rather than a population-wide rate shift.

4.4 Training profiles and held-out scoring

For a target query (e^*, s^*), training environments were selected without using the neural response of e^*. At a source seam s, the wall and open profiles were

$$\begin{aligned} \boldsymbol{\mu}_{W,s} &= \mathrm{mean}_{e \in \mathcal{T}: z_{e,s}=1}\, \boldsymbol{\delta}_{e,s}, \\ \boldsymbol{\mu}_{O,s} &= \mathrm{mean}_{e \in \mathcal{T}: z_{e,s}=0}\, \boldsymbol{\delta}_{e,s}. \end{aligned}$$

computed on the common registered-cell support. The estimated source effect was $\mathbf{d}_s = \boldsymbol{\mu}_{W,s} - \boldsymbol{\mu}_{O,s}$. The held-out outcome was $\boldsymbol{\delta}_{e^*,s^*}$. Scores were Spearman correlations across registered cells. Wall-open advantages compare correlations within the same query and cell support.

The exact-location estimator set $s = s^*$. Its context-matched version restricted training comparisons so the broader internal-wall context was balanced between the wall and open profiles. The reported exact-location benchmark for cross-location prediction was recomputed on the exact queries, sessions, cells, and local supports eligible for the shifted analysis.

4.5 Single-tile counterfactual

The u and o geometries differed at one lattice tile: u blocked tiles $\{4, 5\}$, whereas o blocked only tile $\{4\}$. We formed the cell-wise $u - o$ contrast at the focal seam and evaluated wall and open outcomes at that seam in held-out i, l, and t geometries. Near (2.5–7.5 cm) and far (17.5–22.5 cm) bands were screened separately, so their difference is descriptive. For the order-placebo analysis, we formed later-minus-earlier contrasts at unchanged seams with the same processing and compared their signs with the actual $u - o$ contrast.

4.6 Cross-location prediction

For each eligible target seam, the source seam was a non-overlapping lattice seam 25 cm away with the same signed wall normal and admissible wall/open training support. Source training and target outcome sessions were disjoint. The primary shifted score was

$$\rho_{\mathrm{shift}} = \rho_{\mathrm{S}}\left(\mathbf{d}_{s_{\mathrm{source}}}, \boldsymbol{\delta}_{e^*, s_{\mathrm{target}}}\right).$$

Secondary specificity scores compared the source effect with a training-derived open profile at the target and compared source-wall with source-open predictors. The exact-location benchmark used $\mathbf{d}_{s_{\mathrm{target}}}$ on the identical held-out query support. Stored-map and behavior-adjusted audits agreed on 405 target queries and 658 source pairs, including their session and seam identities and their counts of common cells and bins. That audit did not hash the complete cell-ID or bin-coordinate arrays.

4.7 Behavior-adjusted response estimates

We estimated activity jointly with spatial-bin indicators and nuisance covariates for instantaneous speed, sine and cosine of heading, second heading harmonics, and elapsed time. Spatial effects were evaluated at a common reference covariate distribution calibrated from the first familiar-square session. The no-time sensitivity omitted elapsed time but retained the other covariates. Target-session activity was used to estimate nuisance coefficients for that target map, but target neural outcomes never entered source-profile training.

4.8 Same-session mirror-open control

The mirror-open seam was specified before its neural endpoint was inspected. For each target wall, it lay in the same target session and matched the source-wall pair in translation distance, translation axis, signed normal, relative spatial bins, and registered cells. Both source-to-target nearest-bin distances were 5 cm. Only tangential translations admitted this exact geometric pairing. Primary scoring used globally demeaned source and target vectors. Eligibility required a complete source, wall-target, and open-target triplet on common support.

4.9 Spatial-null calibration

We calibrated two empirical spatial controls at the animal level. The first compared correctly oriented source seams with incorrectly oriented sources at the same 25 cm target-midpoint distance. A tangential stratum additionally matched translation axis and the 5 cm nearest-strip lag. The second compared the wall target with the same-session reflected-open target described above. Both controls retain the observed neural maps, including their place-field smoothness. For each paired animal-level contrast, we enumerated all 2^N sign assignments and calculated the fraction of null means at least as large as the observed mean. These are descriptive exact sign-flip calibrations, not confirmatory population p-values.

4.10 Cell-identity permutation

For each animal, we generated 999 coherent random mappings of source-cell identities while preserving query geometry, target vectors, support sizes, and the mapping across all queries in that permutation. The descriptive upper-tail fraction was $(1+n_{\text{null}\geq\text{observed}})/(1000)$. Correlations and registration constraints limit exchangeability; the fractions were used as identity-specificity diagnostics rather than inferential population p-values.

4.11 Direction, composition, and experience analyses

Direction controls compared eligible source-target pairs with same or opposite signed wall normals while documenting translation axis, distance, overlap, and layout matching. No single comparison simultaneously matched all of these features across the full cohort.

For composition, we identified complete wall-state factorials in which two single-wall effects and their joint state could be estimated. The additive prediction was the sum of the two single effects. It was compared with an oracle best-single-component predictor using correlation and normalized prediction error. Only two repetitive factorials and five eligible animals were available.

Experience analyses contrasted transfer changes across exposure blocks using difference-in-differences variants. Because geometry, order, and exposure were not independently randomized, these estimates were treated as exploratory sensitivity analyses and not as evidence for learning.

4.12 Independent former-barrier analysis

We used the public neutral-barrier data from Blair et al. (2023b) to test whether the former barrier center retained an excess neural similarity relative to matched control locations after removal. The primary endpoint was a Pearson correlation among active cells, averaged as a former-barrier center-minus-control contrast. Its definition was written before the neural endpoint was run in the local project workflow. Spearman scoring and leave-one-animal-out summaries were sensitivities.

4.13 Statistical reporting and reproducibility

Animals were treated as the biological units. We report animal-level means, sign counts, query counts, and explicitly labeled descriptive permutation fractions. No cell-level or query-level observation was presented as an independent animal replicate. The analyses are exploratory: neither the original experiment nor the present reanalysis was preregistered for the cross-location hypothesis.

All figure values derive from tracked machine-readable source files. The repository contains scripts, eligibility ledgers, result summaries, design locks, and an inspectable Git history. The principal manuscript figure is generated from its tracked source-data CSV.

5 Supplementary exploratory analyses

5.1 Direction and two-wall composition

Direction controls did not identify a general orientation rule. Same- versus opposite-signed normals for tangential translation favored the same direction in three of six eligible animals. A normal-axis comparison against opposite-facing or overlapping alternatives was positive in four of seven. A less controlled same- versus opposite-away comparison was positive in all seven, but it also changed strip distance and layout. These analyses support transfer between selected corresponding seams, not rotation- or direction-invariant transport.

Only two complete two-wall factorials were available, both involving the same bit and donut layouts. In five eligible animals, the demeaned additive predictor reached mean correlation 0.153, below an oracle best-single-component predictor at 0.178. The additive advantage was -0.024 and positive in one of five animals. This sparse support provides no positive evidence for linear two-wall composition but cannot establish general non-compositionality.

5.2 Experience and former-barrier endpoints

A transfer-by-experience difference-in-differences estimate was positive under one stored-map specification (mean 0.045, four of four eligible animals), but raw-rate and stricter sensitivity analyses were unstable. Environment order, exposure number, and geometry were not independently varied, so the data do not identify experience-dependent stabilization or innateness.

In an independent neutral-barrier dataset (Blair et al., 2023b), the locked Pearson-active former-barrier center-minus-control endpoint was negative in all leave-one-animal-out summaries (animal mean −0.183; one of six animals positive). A Spearman sensitivity was near zero (mean 0.001; three of six positive). This endpoint showed no persistent local representational scar.

Data and code availability

The CA1 source data are available through the archive cited by Lee et al. (2024). The independent miniscope data are available through the archive cited by Blair et al. (2023a). Analysis code, derived source tables, audit reports, and manuscript sources are publicly available at `https://github.com/asim-v/ca1-wall-update-transfer`. The public release excludes the original raw recordings, which remain available from their source archives.

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Author contributions

J.E.B.S. conceived the study, developed and performed the analyses, interpreted the results, prepared the visualizations, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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